ALICE L WALTON SCHOOL OF MEDICINE

Pharmacokinetics–1 (3 of 3) pH and pKa

Last updated May 12, 2026 · 9 min read CC BY-SA 4.0

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Learning Objectives (1)

After completing this brick, you will be able to:

1 Predict impact on drug absorption and/or reabsorption based on pH and pKa values of drugs and explain the clinical relevance.

CASE CONNECTION

TY is a 45-year-old man diagnosed with osteosarcoma and undergoing a chemotherapy regimen with methotrexate. He is anxious about the disease progression, his chemotherapy, and its outcome. You explain to TY that you are going to start him on methotrexate. You inform him that you will also be ordering 1 L IV sodium bicarbonate that has to be initiated 4 h after therapy. TY is not happy to hear more therapy is being added.

He asks, "Why are we adding more drugs now? Why do I need to get a liter of bicarbonate? Can I not just drink water?" What would be your response to TY? Why does TY require the additional sodium bicarbonate? Consider your answer as you read, and we'll revisit VF at the end of the brick.

GO TO CONCLUSION ↓

We have discussed the ADME (Absorption, Distribution, Metabolism, Excretion) processes that encompass pharmacokinetics. In this brick we are going to go into further detail on the absorption/distribution process based on acid-base chemistry. (**You should review concepts of acid-base chemistry from BOOM**)

In diffusion, a drug will passively move from an area of higher concentration to an area of lower concentration. But how do drugs become concentrated on one side of a membrane to begin with? If Fick's law is followed (**refer to physiology brick**), then drugs would always move from higher to lower concentrations in the body. However, this is not the case because movement or distribution of a drug in the body depends on several factors. One important factor is the drug's ionization state.

Most drugs are either a weak base (proton acceptor) or a weak acid (proton donor). Recall from B he acid-dissociation equilibrium constant Ka for that acid.

Hence, when a drug is dissolved in water, it can either give up a proton and become ionized (acid) or accept a proton and become ionized (base).

If HA was an acid, HA would dissociate as : [HA]⇌ [H+] + [A-]

If HB was a base, HB would dissociate as : [HB+]⇌ [H+] + [B]

Remember that the lipid bilayer is composed of fatty acids and they form a barrier. Thus, substances that are lipid-like move through the membrane. The **ionized** form of the drug has an electric charge and cannot cross a lipid membrane, while the **unionized** form of the drug has an easier time going through the lipid bilayer membrane. Thus, an ionized form of the drug is **hydrophilic** (water-loving or lipophobic), and the unionized form of the drug is more **lipophilic** (lipid-loving or hydrophobic).

The same concept can be discussed in terms of the drugs polarity. **Nonpolar drugs** are uncharged and have an easy time passing through the lipid bilayer. Hence, nonpolar drugs are **lipophilic**. On the other hand, **polar drugs** are evenly charged substances and do not pass through the lipid bilayer. Hence, polar drugs are more **hydrophilic**. (Note: In the next brick, we are going to discuss drug metabolism, and this concept is going to be very important to remember)

What form of a drug can cross the lipid bilayer membrane more easily?

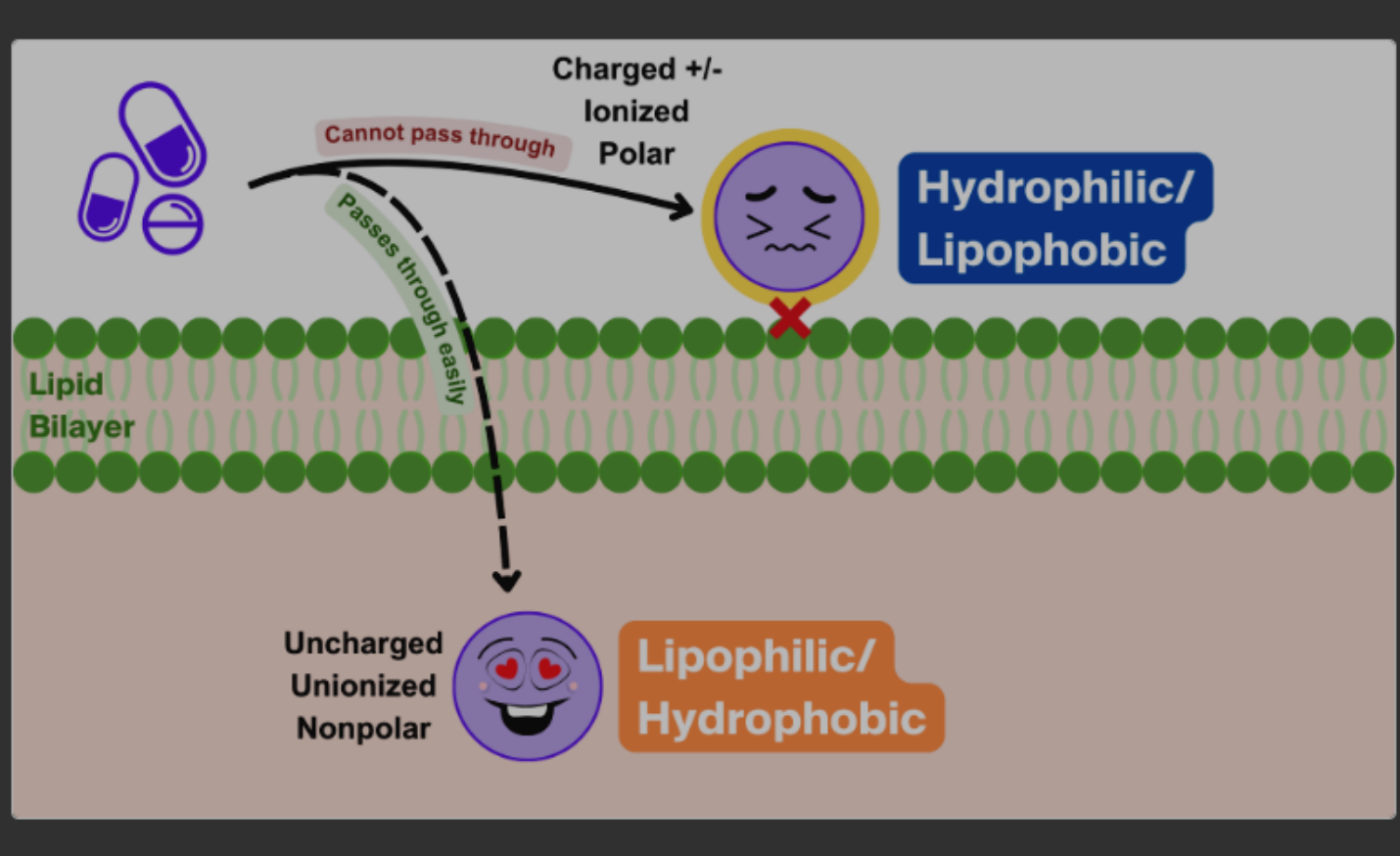


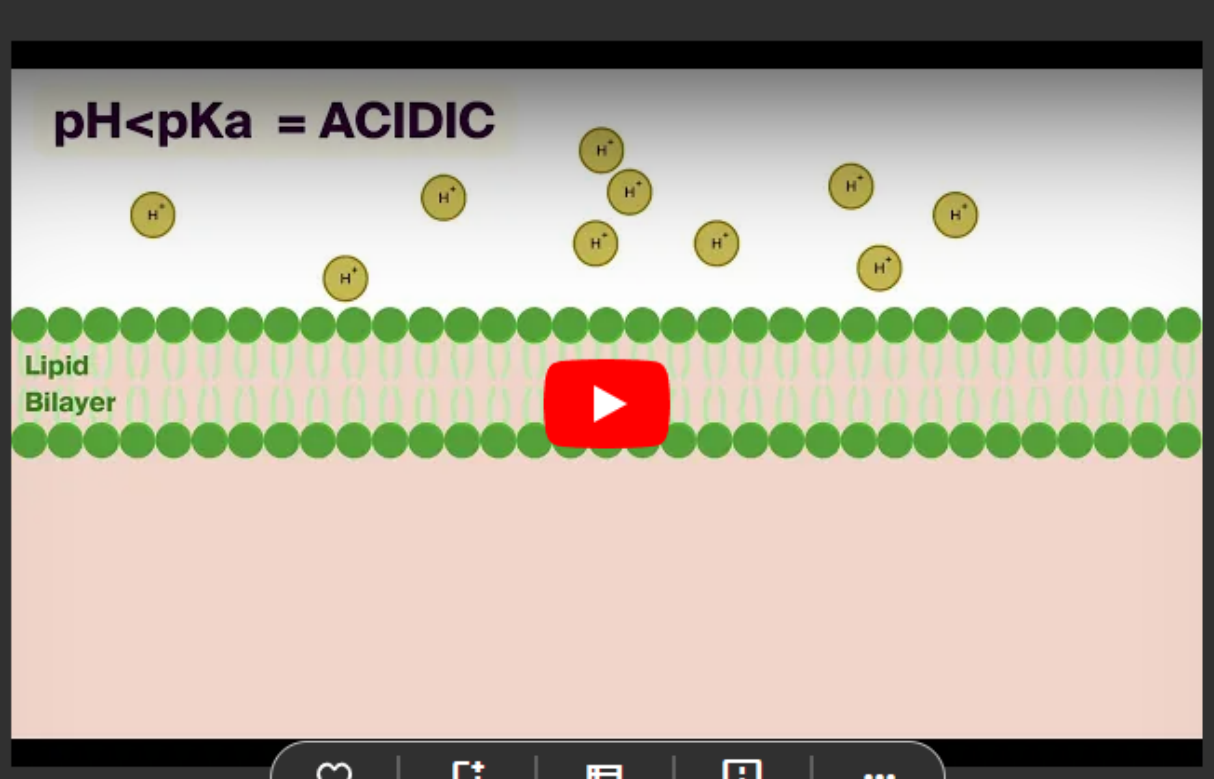
Figure 1

In a solution, the unionized form is in equilibrium with the ionized form. Hence, the next question is how much the ionized form changes. This change depends on two factors: the pH of the surrounding solution and the drug's pKa.

pH of the surrounding solution: Recall that the pH of the solution indicates the [H+] concentration. The lower the pH, the higher the H+ ions. Conversely, the higher the pH, the lower the H+. Since most drugs are weak acids or bases, the rate at which a weak acid or base dissociates depends on the [H^+] concentration. In an alkaline (or basic) environment, weak acids will dissociate to be ionized (giving up H^+). Conversely, in an acidic environment weak base will become ionized (gain H^+).

pKa: The other factor that influences ionization is pka. pKa is the pH at which 50% of the drug is ionized and 50% unionized. There are a few basic rules of thumb that are followed when determining the relationship between pH and pKa that are important to note for drug absorption.

For drugs that are weak acids: [H+] + [A-]→[HA] (HA= non ionized form), A- = ionized form of the weak acid)



Conversely, if the weak acid is placed in an alkaline environment (pH>pKa), then the drug is ionized and is not absorbed (more hydrophilic). In such cases, the drug remains in vascular compartments and interstitial fluid compartments.

For a drug that is a weak acid, if the pH<pKa, then it is likely to cross membranes.

For drugs that are weak bases: [HB^+]→ [H^+] + [B] (weak bases = accepting protons) (BH+ = ionized form of weak base), B= unionized form of weak base.

pH>pKa = ALKALINE
Lipid
Bilayer

On the other hand, if a weak base [HB+] is placed in an acidic environment (pH<pKa), then the drug will remain in its ionized form and is not absorbed (more hydrophilic). In such cases, the drug remains in the vascular compartments and interstitial fluid compartments.

Thus, weakly acidic drugs are better absorbed in an acidic environment, such as the proximal duodenum (near the antral end of the stomach). Conversely, drugs that are weakly basic are absorbed in an alkaline environment, such as the distal ileum.

Figure 2 summarizes the distribution of drugs between ionized and unionized forms of the drug (assume that pKa=6.5).

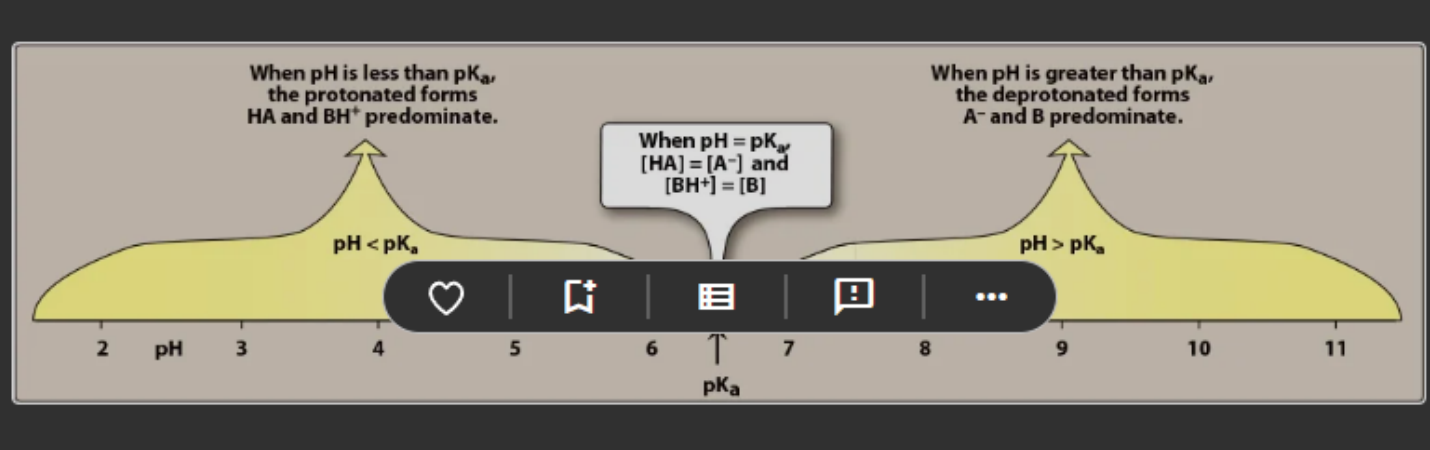


Figure 2

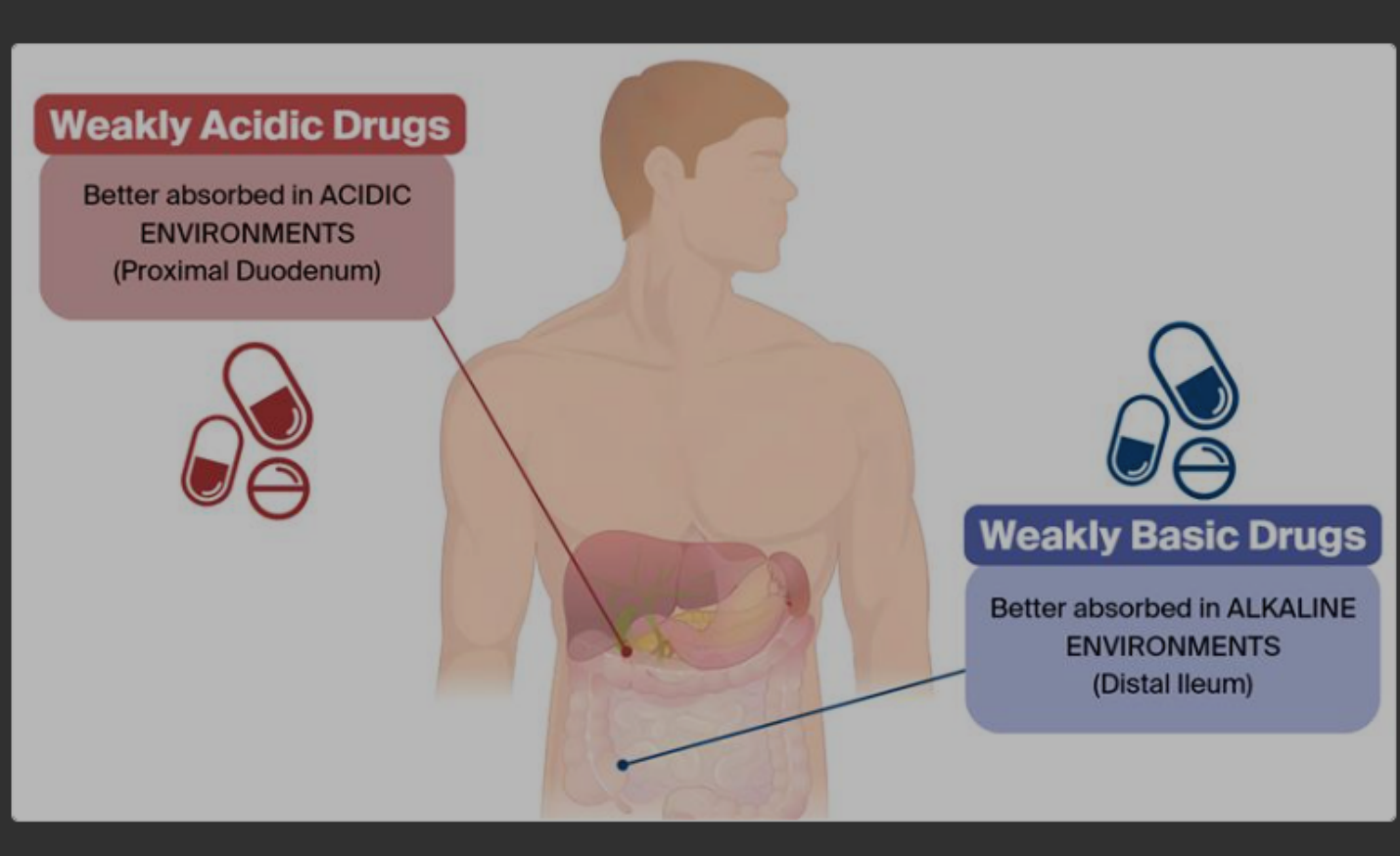

Weakly Acidic Drugs
Better absorbed in ACIDIC ENVIRONMENTS (Proximal Duodenum)
Weakly Basic Drugs
Better absorbed in ALKALINE ENVIRONMENTS (Distal Ileum)

Figure 2

Where are weakly basic drugs best absorbed?

CLINICAL CORRELATION

Despite its acidic environment, the stomach is not ideal for drug absorption, even for weak acids. This is due to its thick mucus layer and limited surface area. The stomach primarily functions as a storage organ rather than one for absorption. In contrast, the small intestine, with its extensive villi and microvilli, offers a large surface area for absorption. Consequently, acidic drugs are typically absorbed in the acidic regions of the proximal duodenum, while basic drugs are better absorbed in the more alkaline areas of the distal ileum.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TY, why is it necessary to add sodium bicarbonate to the methotrexate for his chemotherapy.

You reply to TY that methotrexate is an acid by way of its structure. Sodium bicarbonate is a base or is alkaline in nature. The addition of sodium bicarbonate or base to methotrexate, the acid causes a reaction that leaves methotrexate with a charge. You explain to TY that when something is charged or ionized, it tends to dissolve or stay in water. So, by adding the sodium bicarbonate, the methotrexate 'stays' in water. This helps to 'get rid' of the drug as the kidneys can quickly excrete it. The excretion of m tial adverse effects.

Summary

- Most drugs are either weak acids (proton donors) or weak bases (proton acceptors), and their ionization state critically determines their ability to cross lipid bilayer membranes.
- The unionized (lipophilic) form of a drug crosses lipid membranes more easily, while the ionized (hydrophilic) form remains in vascular and interstitial fluid compartments.
- Drug ionization is governed by two key factors: the pH of the surrounding solution and the drug's pKa, which is the pH at which 50% of the drug is ionized and 50% is unionized.
- Weak acids are better absorbed in acidic environments (pH < pKa), such as the proximal duodenum, while weak bases are better absorbed in alkaline environments (pH > pKa), such as the distal ileum.
- Despite its acidic environment, the stomach is not ideal for drug absorption due to its thick mucus layer and limited surface area; the small intestine, with its large surface area from villi and microvilli, is the primary site of drug absorption.

Review Questions

1. Which form of a drug can easily cross a lipid membrane?

A. ○ Ionized
B. ● Unionized
C. ○ Both forms

Hide Explanation

The correct answer is unionized form (B). The unionized form of the drug is more lipophilic (B), allowing it to pass through the lipid bilayer membrane. The ionized form (A) is hydrophilic and cannot cross the lipid membrane easily. Both forms (C) is incorrect because only the unionized form can cross easily.

2. If a drug is a weak acid, it is most likely to get absorbed in which of these locations?

A. ○ distal ileum
B. ● proximal duodenum
C. ○ late duodenum
D. ○ late ileum

Hide Explanation

The correct answer is the proximal duodenum(B). Given its proximity to the stomach, the acidic s nized. The unionized form of the drug is lipophilic and, hence, is absorbed. Choice (A) is incorrect, as the environment in the distal ileum is more alkaline (or basic). Since the late duodenum is farther from the stomach, it is more alkaline (or basic); hence, Choice (C) is incorrect since the surroundings are more alkaline (or basic). Late ileum, Choice (D), is more basic surroundings; hence the answer is incorrect.

3. If a drug has a weak base and is placed in an alkaline environment, the drug is most likely to

A. ○ cross into interstitial space
B. ● cross the cell membrane
C. ○ remain in the vascular compartment

Hide Explanation

The correct answer is 'cross the cell membrane' (B). Since the drug is a weak base in an alkaline environment with fewer protons, it will give up its proton to become its unionized (or uncharged) form. Thus, the unionized form of the drug is more likely to cross the cell membrane. Choice (A) and (C) are incorrect as the unionized form of the drug is not hydrophilic. Ionized form of the drug is more hydrophilic and tend to remain in the interstitial space or the vascular compartment.

How would you rate this Brick?

Hide Explanation

The correct answer is 'cross the cell membrane' (B). Since the drug is a weak base in an alkaline environment with fewer protons, it will give up its proton to become its unionized (or uncharged) form. Thus, the unionized form of the drug is more likely to cross the cell membrane. Choice (A) and (C) are incorrect as the unionized form of the drug is not hydrophilic. Ionized form of the drug is more hydrophilic and tend to remain in the interstitial space or the vascular compartment.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Predict impact on drug absorption and/or reabsorption based on pH and pKa values of drugs and explain the clinical relevance.~~

Objective Confidence: 1/1